

Report on YOLO Dataset Meta Files

YOLO Dataset Meta Files Report (YOLOv26)

Introduction

YOLO (You Only Look Once) models use structured datasets along with meta configuration files to perform object detection and segmentation. This report analyzes the dataset configuration file `coco8-seg.yaml` and corresponding label files to understand dataset organization and annotation format.

Dataset Configuration File (YAML)

The YAML file acts as the primary meta file and defines dataset structure and class information.

Key Details:

- **Training and Validation Paths:**

```
train: images/train
val: images/val
```

- **Class Information:**

The YAML file maps numeric class IDs to object names from **0 to 79**, indicating **80 object classes**.

The YAML file specifies:

- dataset location
 - training and validation paths
 - class IDs and corresponding object names
-

Directory Structure

```
coco8-seg/
├── images/
│   ├── train/
│   └── val/
├── labels/
│   ├── train/
│   └── val/
```

- `images/` → input images
 - `labels/` → annotation files
 - Each image has a corresponding `.txt` label file
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Label Files Analysis

Each label file contains multiple lines, where each line represents one object.

Format Observed:

```
45 0.782016 0.986521 0.937078 ...
```

Interpretation:

- First value → **class ID**
- Remaining values → **coordinates**

Key Insight:

- Multiple coordinate values indicate **polygon-based annotations**
- This confirms the dataset is used for **segmentation**
- Coordinates are **normalized (0 to 1)** relative to image dimensions

My Observation

From the analysis:

- The YAML file contains **class numbers mapped to object names**
- There are **80 object classes (0–79)**
- Label files contain **class IDs followed by coordinate values**
- The large number of coordinates per object shows **segmentation masks instead of simple bounding boxes**

Conclusion

YOLO dataset meta files define how data is structured and interpreted during training. The YAML file provides dataset paths and class mappings, while label files contain detailed object annotations. In this dataset, segmentation is performed using polygon coordinates. Although referred to as YOLOv26, the dataset format follows standard YOLO conventions used in modern implementations.

References

- Ultralytics Dataset Documentation: <https://docs.ultralytics.com/datasets/>
- COCO8-Seg Dataset: <https://docs.ultralytics.com/datasets/segment/coco8-seg/>
- Ultralytics Repository: <https://github.com/ultralytics/ultralytics>